"Harrier 3.0 TD: Engine over-temperature caused by failing water pump seal"

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1 Condition

Owners of Harrier 3.0 TD vehicles may report an engine over-temperature condition accompanied by an illuminated check engine lamp. On scan, the powertrain control module stores DTC P0217 (Engine Over-Temperature Condition), often after sustained highway operation or towing. Coolant loss may be evident around the front of the engine, and the temperature gauge may read into the red zone before the warning is acknowledged. For background on how the regulating system should behave, see the Harrier Cooling Service Manual :: Coolant Circulation .

2 Cause

The root cause is a degraded mechanical seal on water pump assembly WP-6122, which permits coolant seepage and progressive loss of circulation, ultimately setting P0217. In a subset of cases the original thermostat TSTAT-6100 is found stuck or sticking, compounding the over-temperature event, though the pump seal is the primary failure. Technicians should not condemn the thermostat without confirming pump integrity first; refer to Harrier Cooling Service Manual :: Water Pump Identification and Supersession to verify the installed part level. Diagnosis of the temperature regulation path is covered in Harrier Cooling Service Manual :: Thermostat and Temperature Faults .

3 Repair Procedure

Perform the water pump replacement following procedure PROC-WATER-PUMP, replacing pump assembly WP-6122 with the latest superseding part and a new gasket. Drain, refill, and bleed the system per the full Harrier Cooling Service Manual :: Water Pump Replacement Procedure , then clear P0217 and road test to confirm the temperature stabilizes. If the thermostat is also being serviced, complete the Harrier Cooling Service Manual :: Thermostat Replacement Procedure in the same visit. For related engine-side inspection of the seal area, consult the Harrier Engine Service Manual .